

Notes: Cronqvist, Ladika, Pazaj, and Sautner (JFE, 2024)

Limited Attention to Detail in Financial Markets: Evidence from Reduced-Form and Structural Estimation

Contents

1	Big picture	3
2	Data and measurement	3
2.1	FAS 123-R as a change in visibility	3
2.2	High-option and low-option firms	4
2.3	Main outcome variables	4
2.4	Timing conventions	4
2.5	Confounding-channel variables	4
3	Models and empirical specifications	5
3.1	Baseline staggered difference-in-differences	5
3.2	Quarter-by-quarter learning specification	5
3.3	Reduced-form tests for confounding channels	5
3.4	Structural model: core objects	6
3.5	Structural estimation and counterfactuals	6
4	Identification and empirical design	6
4.1	What identifies the limited-attention effect	6
4.2	Why fundamentals are not the main story	6
4.3	The compound exclusion restriction problem	7
4.4	Why analysts and investors are analyzed separately	7
4.5	Limitations	7
5	Tables and figures	8
5.1	Figure 1 + Figure 2: timing of hypotheses and staggered compliance	8
5.2	Table 1: summary statistics and pre-trends	8
5.3	Table 2: FAS 123-R and reported earnings	8
5.4	Table 3: analyst missed forecasts	9
5.5	Table 4: analyst recommendations	10
5.6	Table 5: stock market valuations	10
5.7	Table 6 + Table 7: confounding channels and controls	11
5.8	Structural model panel + Table 8: SMM estimation and counterfactuals	12
5.9	Table 9 + Table 10: analyst learning	13
5.10	Table 11 + Table 12: investor learning and option-related return dynamics	14
6	Short summary	15
7	Conclusion	15
7.1	Core conclusion	15

7.2	Economic intuition	15
7.3	Methodological takeaway	15
7.4	Takeaway	15

1 Big picture

- **Question.** Does the way information is presented in financial statements affect how sophisticated market participants process value-relevant information?
- **Setting.** The paper studies FAS 123-R, which required U.S. firms to expense stock option compensation in the income statement beginning in 2005–2006. Before the rule, firms disclosed option costs in footnotes.
- **Core idea.** FAS 123-R changed the visibility of option compensation costs, not the economic magnitude of the costs. This makes it useful for studying limited attention to detail.
- **Key treatment variation.** Compliance timing was staggered because each firm complied in the first new fiscal year beginning after June 2005. Cross-sectional intensity came from the firm’s pre-rule option pay.
- **Main reduced-form result.** High-option firms reported lower EPS after FAS 123-R and were more likely to miss analyst EPS forecasts. The effect is strongest in the first two fiscal quarters after compliance and then fades, consistent with analyst learning.
- **Analyst behavior.** Analysts downgraded high-option firms after compliance. Recommendations remained lower, suggesting analysts learned faster than investors that high-option firms had previously been overvalued.
- **Investor behavior.** High-option firms’ market-to-book ratios fell after FAS 123-R. The decline widened across fiscal quarters, suggesting investors only gradually learned to incorporate option costs.
- **Methodological challenge.** FAS 123-R has been reused in many papers and affected several outcomes besides disclosure visibility. Reusing the natural experiment requires addressing compound exclusion restriction concerns.
- **Methodological contribution.** The paper combines reduced-form tests of confounding channels with a structural model that quantifies limited attention before and after the accounting change.
- **Structural result.** Estimated limited attention is large in the non-expensing regime, smaller in the interim regime, and small in the expensing regime. The model implies that most of the valuation drop is due to learning about option costs.
- **Economic takeaway.** Even analysts and investors can underweight information when it is buried in less salient reporting formats. Moving the same information from footnotes to the income statement can change forecasts, recommendations, and firm valuations.

2 Data and measurement

2.1 FAS 123-R as a change in visibility

- Before FAS 123-R, firms had to disclose the fair value of stock option compensation in footnotes, but the costs generally did not reduce reported income-statement earnings.
- FAS 123-R required firms to deduct option grant costs from earnings, starting with the first new fiscal year beginning after June 2005.
- The rule affected the presentation of already available information: option cost information existed before the rule, but was less salient.
- Firms with greater historical option compensation were more exposed to the new reporting rule.

Interpretation: The paper treats FAS 123-R as an increase in information visibility. This is distinct from a shock to fundamentals because the accounting rule changed the location of information, not firms’ actual option grants or operating revenues.

2.2 High-option and low-option firms

- The treatment intensity is based on pre-FAS 123-R option pay.
- A firm is classified as a high-option firm if the fair value of options granted divided by total assets is above the sample median in fiscal year 2004 or fiscal year 2005.
- Low-option firms are below the sample median.
- The key regression variation is the interaction of the post-FAS 123-R indicator with the high-option firm indicator.

Interpretation: The design compares high-option and low-option firms before and after compliance, using high-option firms as the group for which option expense visibility should matter most.

2.3 Main outcome variables

- **Earnings outcomes:** GAAP EPS and EBIT/share.
- **Forecast outcome:** Missed Forecast equals one if reported quarterly EPS fall below the immediately preceding consensus analyst forecast.
- **Recommendation outcomes:** Analyst Rec, Pct. Buy Rec, and Pct. Sell Rec from IBES recommendation data.
- **Valuation outcome:** Market-to-Book Ratio at the end of each fiscal quarter.
- **Earnings announcement outcome:** EA CAR, the cumulative abnormal return around earnings announcements.

Interpretation: The outcomes trace the learning chain: accounting visibility first changes reported earnings, then affects analysts' forecasts and recommendations, and finally changes investor valuations.

2.4 Timing conventions

- FQ1 is the first fiscal quarter in which FAS 123-R affects reported EPS.
- FQ2 is when investors observe the FQ1 earnings announcement and forecast miss information.
- The main window compares fiscal quarters $[-4, +4]$ around each firm's staggered FAS 123-R compliance date.
- The paper measures forecast misses, recommendations, and market valuations at precise points in the fiscal-quarter timeline.

Interpretation: The paper's timing is important because limited attention predicts gradual learning. Analysts and investors should not necessarily incorporate the new accounting treatment immediately at the compliance date.

2.5 Confounding-channel variables

- The authors review prior FAS 123-R papers and identify many additional outcomes affected by the regulation.
- They group possible confounds into three channels: risk-taking, manipulation, and CEO turnover.
- Risk-taking variables include volatility, acquisitions, leverage, capex, and related firm policies.

- Manipulation variables include repurchases, discretionary accruals, earnings management, and related reporting choices.
- CEO turnover is examined because option acceleration and compensation changes could affect executives.

Interpretation: This part of the paper is a guide for reusing natural experiments. When one policy shock has multiple effects, researchers must show that the alternative channels do not drive the outcome of interest.

3 Models and empirical specifications

3.1 Baseline staggered difference-in-differences

The main reduced-form model is:

$$Y_{f,t} = \alpha_f + \delta_t + \beta (PostFAS123R_{f,t} \times HighOptionFirm_f) + \gamma' X_{f,t} + \varepsilon_{f,t}. \quad (1)$$

- $Y_{f,t}$ is an earnings, analyst, valuation, or return outcome.
- $PostFAS123R_{f,t}$ equals one after firm f begins complying with FAS 123-R.
- $HighOptionFirm_f$ equals one for firms with high pre-rule option use.
- Firm fixed effects absorb time-invariant differences across firms.
- Calendar-quarter or fiscal-quarter fixed effects absorb aggregate shocks and common trends.

Interpretation: A positive coefficient for Missed Forecast or a negative coefficient for Market-to-Book Ratio means high-option firms changed more after FAS 123-R than low-option firms did.

3.2 Quarter-by-quarter learning specification

The paper also decomposes the effect by post-compliance quarter:

$$Y_{f,t} = \alpha_f + \delta_t + \sum_{q=1}^4 \beta_q (PostFAS123R_{FQq,t} \times HighOptionFirm_f) + \gamma' X_{f,t} + \varepsilon_{f,t}. \quad (2)$$

Interpretation: If analysts learn, the missed-forecast coefficient should be strongest in FQ1 and FQ2 and then disappear. If investors learn slowly, valuation effects can widen across later quarters.

3.3 Reduced-form tests for confounding channels

To evaluate possible violations of the compound exclusion restriction, the authors estimate:

$$Z_{f,t} = \alpha_f + \delta_t + \theta (PostFAS123R_{f,t} \times HighOptionFirm_f) + \gamma' X_{f,t} + u_{f,t}, \quad (3)$$

where $Z_{f,t}$ is each potential confounding outcome identified in the literature.

Interpretation: If a candidate confounding variable does not change for high-option firms relative to low-option firms, it is unlikely to explain the main analyst and valuation results. If it does change, the authors add it as a control in the main regressions.

3.4 Structural model: core objects

The paper builds a dynamic principal-agent model that extends Marinovic and Varas (2019). Firm value is the discounted value of cash flows net of CEO compensation. The CEO can manipulate current cash flows and choose risk-taking incentives; manipulation accumulates and lowers future cash flows.

The key attention parameters are:

$$\alpha_N, \quad \alpha_{E^-}, \quad \alpha_E, \quad (4)$$

where N is the non-expensing regime, E^- is the interim regime between announcement and implementation, and E is the expensing regime.

Interpretation: Lower attention means the market underweights compensation costs. The structural model estimates how the market’s attention to option costs changes across regimes.

3.5 Structural estimation and counterfactuals

The structural model is estimated by simulated method of moments (SMM). Moments include reduced-form effects on valuation, compensation, and policy outcomes. The counterfactual exercises shut down specific channels:

- limited attention changes;
- manipulation incentives;
- risk-taking incentives.

Interpretation: The structural model answers a question the reduced-form tests cannot quantify: how much of the valuation decline is due to learning about option costs rather than simultaneous changes in risk-taking or manipulation incentives.

4 Identification and empirical design

4.1 What identifies the limited-attention effect

- Staggered compliance creates time-series variation: firms with different fiscal year ends begin expensing options at different times.
- High-option versus low-option firms create cross-sectional exposure variation.
- Analyst and valuation outcomes are examined before and after compliance within the same firm.
- The paper checks pre-trends before FAS 123-R to support the parallel-trends assumption.

Interpretation: Identification comes from the interaction of exposure and timing. High-option firms should be affected more strongly by the same visibility shock.

4.2 Why fundamentals are not the main story

- FAS 123-R changed the accounting presentation of option costs, not the operating fundamentals of the firm.
- High-option firms’ relative earnings decline comes from option expense recognition.
- The paper reports no comparable deterioration in revenues.
- Analyst misses and recommendations track the new visibility of option costs rather than a fundamental decline in operations.

Interpretation: The fundamental-performance channel would imply deteriorating revenues or operating performance. The evidence instead points to a reporting-salience effect.

4.3 The compound exclusion restriction problem

- FAS 123-R has been used in many empirical papers, and the regulation changed multiple firm outcomes.
- Reusing the natural experiment requires assuming that these additional outcomes do not drive the new paper's dependent variables.
- Each additional effect of the regulation creates another exclusion restriction.
- The paper explicitly maps the possible confounding channels and tests them.

Interpretation: This is the methodological contribution. The paper does not simply assert the regulation is clean; it demonstrates how to audit a reused natural experiment.

4.4 Why analysts and investors are analyzed separately

- Analysts are sophisticated but cognitively constrained information processors.
- Investors can learn from analysts' forecasts, recommendation changes, and earnings-announcement surprises.
- The paper finds analysts learn within a few quarters, while investors' valuation adjustments occur more gradually.

Interpretation: The separate analysis shows a dynamic information diffusion process: analysts update first, and market valuations adjust more slowly.

4.5 Limitations

- FAS 123-R still affected compensation design and other firm choices, so no reduced-form design can eliminate every concern.
- Structural estimates depend on model assumptions.
- Option cost visibility is measured through high-option exposure rather than directly observing each investor's attention.
- The setting studies option compensation and accounting visibility; generalizing to all disclosure settings requires caution.

Interpretation: The paper is strongest as evidence that presentation format matters for sophisticated agents, and as a methodological template for combining reduced-form and structural methods.

5 Tables and figures

5.1 Figure 1 + Figure 2: timing of hypotheses and staggered compliance

Read:

- Figure 1 lays out the sequence: FAS 123-R takes effect, analysts forecast FQ1 EPS, firms announce FQ1 EPS, and investors update valuations by FQ2.
- The paper measures Missed Forecast and EA CAR at the FQ1 announcement, Analyst Rec before the FQ1 announcement, and Market-to-Book Ratio at the end of FQ2.
- Figure 2 shows the staggered compliance rule based on fiscal year ends.
- Firms with fiscal years ending in June complied first; firms with fiscal years ending in May complied last.

Economic message: The empirical design is built around precise timing. The figures show how the paper separates analyst learning, earnings announcements, and investor valuation adjustments.

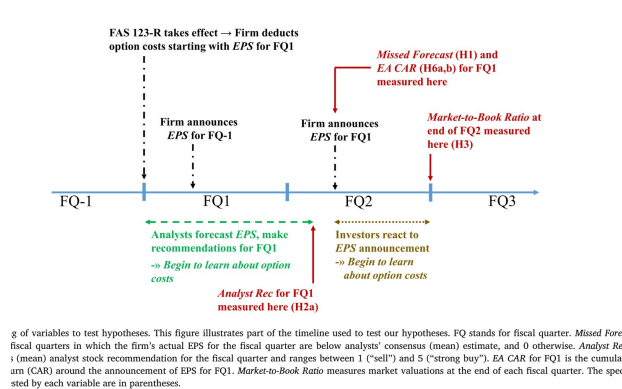


Fig. 1: timing of variables and hypotheses.

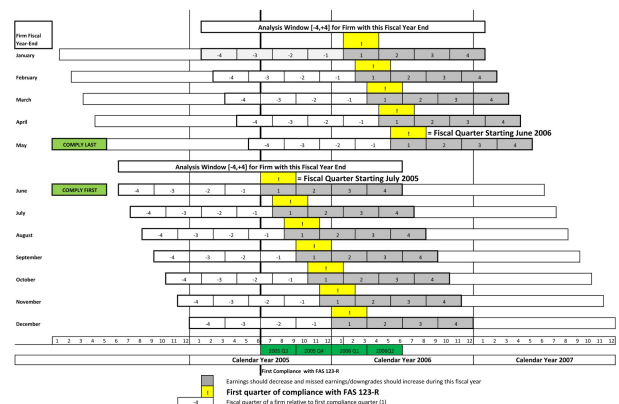


Fig. 2: staggered FAS 123-R compliance schedule.

5.2 Table 1: summary statistics and pre-trends

Read:

- The sample contains 3,906 U.S. firms in Compustat and IBES.
- Average GAAP EPS is 0.131 and median GAAP EPS is 0.070.
- The mean Missed Forecast rate is 37.7%.
- About 54.1% of analysts' recommendations are buys or strong buys, while about 6.0% are sells or strong sells.
- The average market-to-book ratio is 2.363.
- Panel B compares pre-FAS 123-R trends for high-option and low-option firms and finds no large divergent pre-trends in the main outcome variables.

Economic message: Table 1 supports the identifying design by showing that high-option and low-option firms do not display strong divergent trends before the accounting rule becomes binding.

5.3 Table 2: FAS 123-R and reported earnings

Read:

Table 1
Summary statistics.

Panel A: Firm characteristics						
	Mean	Median	25%	75%	St. Dev.	Obs.
GAAP Earnings						
$EPS_{f,j}$	0.131	0.070	-0.060	0.320	0.444	29,878
$EBIT/Share_{f,j}$	0.381	0.170	-0.036	0.625	0.717	28,049
$Sales/Share_{f,j}$	5.191	2.439	0.657	6.733	7.007	28,242
Pro Forma Earnings						
$EPS_{f,j}$	0.059	0.110	-0.050	0.330	0.737	27,050
$EBIT/Share_{f,j}$	0.399	0.214	-0.019	0.658	0.667	22,985
$Sales/Share_{f,j}$	5.378	2.705	0.786	7.054	6.936	26,048
High-Option Firm_f						
$Missed Forecast_{f,j}$	0.377					26,157
$Analyst Rec_{f,j}$	3.739	3.750	3.330	4.100	0.617	19,658
$Pct. Buy Rec_{f,j}$	54.097	50.000	33.330	78.570	31.728	19,454
$Pct. Sell Rec_{f,j}$	6.042	0.000	0.000	6.670	13.656	19,454
$Log Assets_{f,j}$	5.510	5.467	4.083	6.869	1.975	28,864
$Market-to-Book Ratio_{f,j}$	2.363	1.785	1.303	2.724	1.671	27,423
$EA CAR_{f,j}$	-0.002	-0.002	-0.055	0.051	0.112	24,562
$Leverage_{f,j}$	0.198	0.134	0.001	0.315	0.222	28,093
$Investment/Sales_{f,j}$	0.102	0.022	0.007	0.058	0.247	26,537
$Analyst Coverage_{f,j}$	7.001	5.000	3.000	10.000	5.985	19,667
Panel B: Pre-trends in outcome measures before FAS 123-R						
	Low-Option Firms			High-Option Firms		
	Mean	Median	St. Dev.	Mean	Median	St. Dev.
$\Delta GAAP EPS_{f,j}$	0.015	0.010	0.432	0.011	0.010	0.298
$\Delta Pro Forma EPS_{f,j}$	0.013	0.010	0.450	0.012	0.005	0.461
$\Delta Missed Forecast_{f,j}$	0.003	0.000	0.607	0.004	0.000	0.618
$\Delta Analyst Rec_{f,j}$	-0.026	0.000	0.343	-0.034	0.000	0.457
$\Delta Pct. Buy Rec_{f,j}$	-1.292	0.000	19.067	-1.401	0.000	25.694
$\Delta Pct. Sell Rec_{f,j}$	0.490	0.000	10.091	0.335	0.000	11.852
$\Delta Market-to-Book Ratio_{f,j}$	-0.001	0.006	0.415	0.009	0.000	0.864
						-0.010
						(-1.54)

Panel A shows summary statistics for the sample. The sample is 3,906 U.S. firms in the intersection of the Compustat and IBES databases. The sample excludes financials and utilities, firms that voluntarily adopted option expensing, and firms that changed their fiscal year-end in 2005 or 2006. Observations are at the firm-fiscal quarter level, for the four fiscal quarters before to four fiscal quarters after each firm's (staggered) FAS 123-R compliance date. Panel B follows Lemmon and Roberts (2010) and compares the growth rates in the key dependent variables across high- and low-option firms for the fiscal years before FAS 123-R took effect. Observations are at the firm-fiscal quarter level, for the sixteen fiscal quarters before to one fiscal quarter before each firm's (staggered) FAS 123-R compliance date. The panel also presents difference-in-means tests. High-option firms are defined as firms with an above-median ratio of *Fair Value of Options Granted/Total Assets* in either fiscal year 2004 or fiscal year 2005. Low-option firms are all other firms. ***, **, * indicate significance levels of 1%, 5%, and 10%, respectively. Variable definitions are in Data Appendix A.

Table 2
Effect of FAS 123-R on GAAP earnings.

Dependent Variable	$EPS_{f,j}$	$EBIT/Share_{f,j}$	$EPS_{f,j}$	$EBIT/Share_{f,j}$
Estimation Window	[-4;+4] Fiscal Quarters around FAS 123-R			
	(1)	(2)	(3)	(4)
$Post FAS 123-R_{f,j} \times High-Option Firm_f$	-0.046*** (-5.34)	-0.033*** (-2.72)	-0.040*** (-4.98)	-0.021* (-1.83)
$Post FAS 123-R_{f,j}$	0.030*** (4.39)	0.006 (0.62)	0.170** (1.98)	0.254* (1.88)
$High-Option Firm_f$	0.061*** (4.69)	0.100*** (4.84)		
$Log Assets_{f,j-1}$	0.094*** (25.93)	0.178*** (29.23)	-0.024* (-1.82)	0.079*** (4.45)
$Market-to-Book Ratio_{f,j-1}$	0.042*** (15.51)	0.046*** (10.23)	0.023*** (7.60)	0.031*** (8.92)
$Leverage_{f,j-1}$	-0.190*** (-7.32)	0.142*** (3.67)	0.006 (0.15)	0.130*** (2.75)
$Investment/Sales_{f,j-1}$	-0.286*** (-13.81)	-0.370*** (-10.74)	0.033 (1.53)	0.002 (0.08)
Industry Fixed Effects	Yes	Yes	No	No
Fiscal-Year-End Fixed Effects	Yes	Yes	No	No
Firm Fixed Effects	No	No	Yes	Yes
Fiscal-Quarter Fixed Effects	No	No	Yes	Yes
Calendar-Quarter Fixed Effects	No	No	Yes	Yes
Obs.	22,849	22,542	22,849	22,542
Adj. R-sq.	0.281	0.400	0.011	0.024

The sample is 3,906 U.S. firms in the intersection of the Compustat and IBES databases. The sample excludes financials and utilities, firms that voluntarily adopted option expensing, and firms that changed their fiscal year-end in 2005 or 2006. Observations are at the firm-fiscal quarter level, for the four fiscal quarters before to four fiscal quarters after each firm's (stag-

- High-option firms report significantly lower earnings after FAS 123-R.
- Column 1 estimates that EPS falls by 0.046 for high-option firms relative to low-option firms after compliance.
- Column 2 estimates an EBIT/share decline of 0.033.
- Specifications with firm, fiscal-quarter, and calendar-quarter fixed effects deliver similar magnitudes.
- The estimated effect is around 10% of the standard deviation of EPS.

Economic message: The rule mechanically makes option costs visible in reported earnings. This is the first stage for testing whether analysts and investors had previously underweighted those costs.

5.4 Table 3: analyst missed forecasts

Read:

- High-option firms become more likely to miss analyst EPS forecasts after FAS 123-R.
- The main coefficient is about 0.067, meaning a 6.7 percentage point increase relative to low-option firms.
- The effect is large relative to the baseline missed-forecast rate.
- Quarter-by-quarter estimates show that the effect is strongest in FQ1 and FQ2.
- The effect disappears by FQ3 and FQ4.

Economic message: Analysts initially underestimated option expenses after the new rule, but learned to incorporate them within a few quarters.

Table 3

Effect of FAS 123-R on missed earnings forecasts.

Dependent Variable	Missed Forecast _{f,j}			
Estimation Window	[-4;+4] Fiscal Quarters around FAS 123-R			
	(1)	(2)	(3)	(4)
Post FAS 123-R _{f,j} × High-Option Firm _f	0.067*** (3.53)	0.060*** (3.11)		
Post FAS 123-R FQ1 _{f,j} × High-Option Firm _f			0.127*** (4.63)	0.122*** (4.46)
Post FAS 123-R FQ2 _{f,j} × High-Option Firm _f			0.084*** (2.98)	0.067** (2.38)
Post FAS 123-R FQ3 _{f,j} × High-Option Firm _f			0.029 (1.01)	0.023 (0.79)
Post FAS 123-R FQ4 _{f,j} × High-Option Firm _f			0.024 (0.83)	0.018 (0.62)
Post FAS 123-R _{f,j}	-0.003 (-0.31)	-0.988*** (-5.11)		
Post FAS 123-R FQ1 _{f,j}			-0.014 (-0.92)	-1.009*** (-5.81)
Post FAS 123-R FQ2 _{f,j}			-0.025 (-1.51)	-1.036*** (-6.08)
Post FAS 123-R FQ3 _{f,j}			0.012 (0.71)	-1.024*** (-5.92)
Post FAS 123-R FQ4 _{f,j}			0.015 (0.86)	-1.096*** (-6.41)
High-Option Firm _f	-0.019 (-0.99)	-0.019 (-0.98)	-0.019 (-0.98)	-0.019 (-0.98)
Log Assets _{f,j-1}	-0.037*** (-6.10)	0.193*** (5.83)	-0.038*** (-6.12)	0.192*** (5.79)
Market-to-Book Ratio _{f,j-1}	-0.018*** (-3.87)	0.026*** (2.63)	-0.018*** (-3.91)	0.024** (2.37)
Leverage _{f,j-1}	0.127*** (3.66)	-0.134 (-1.51)	0.128*** (3.68)	-0.124 (-1.40)
Investment/Sales _{f,j-1}	0.062* (1.68)	-0.117 (-1.35)	0.061* (1.65)	-0.123 (-1.40)
Analyst Coverage _{f,j}	0.000 (0.27)	0.003 (0.99)	0.000 (0.31)	0.003 (1.11)
Industry Fixed Effects	Yes	No	Yes	No
Fiscal-Year-End Fixed Effects	Yes	No	Yes	No
Firm Fixed Effects	No	Yes	No	Yes
Fiscal-Quarter Fixed Effects	No	Yes	No	Yes

Table 4

Effect of FAS 123-R on analyst recommendations.

Dependent Variable	Analyst Rec _{f,j}		Pct. Buy Rec _{f,j}		Pct. Sell Rec _{f,j}	
Estimation Window	[-4;+4] Fiscal Quarters around FAS 123-R					
	(1)	(2)	(3)	(4)	(5)	(6)
Post FAS 123-R FQ1 _{f,j} × High-Option Firm _f	-0.060*** (-3.13)	-0.069*** (-3.73)	-2.650*** (-2.62)	-3.349*** (-3.40)	1.100** (2.51)	1.387*** (3.18)
Post FAS 123-R FQ2 _{f,j} × High-Option Firm _f	-0.089*** (-4.04)	-0.097*** (-4.45)	-4.238*** (-3.55)	-5.032*** (-4.31)	1.657*** (3.29)	1.862*** (3.67)
Post FAS 123-R FQ3 _{f,j} × High-Option Firm _f	-0.065*** (-2.65)	-0.071*** (-2.98)	-3.238** (-2.46)	-4.138*** (-3.21)	1.340** (2.46)	1.476*** (2.68)
Post FAS 123-R FQ4 _{f,j} × High-Option Firm _f	-0.053** (-2.02)	-0.065** (-2.52)	-2.929** (-2.07)	-4.039*** (-2.90)	1.293** (2.16)	1.445** (2.38)
Post FAS 123-R FQ1 _{f,j}	-0.011 (-1.13)	0.668** (2.27)	-1.050** (-2.09)	32.105*** (4.75)	-0.315 (-1.16)	-11.434 (-0.84)
Post FAS 123-R FQ2 _{f,j}	0.020* (1.66)	0.666** (2.27)	0.841 (1.34)	32.463*** (4.85)	-0.755** (-2.26)	-11.266 (-0.83)
Post FAS 123-R FQ3 _{f,j}	0.007 (0.53)	0.649** (2.22)	0.246 (0.36)	31.845*** (4.76)	-0.667* (-1.85)	-11.277 (-0.83)
Post FAS 123-R FQ4 _{f,j}	-0.041*** (-2.86)	0.652** (2.23)	-1.969*** (-2.61)	31.775*** (4.87)	-0.167 (-0.43)	-12.139 (-0.89)
High-Option Firm _f	0.158*** (4.83)	0.158*** (5.03)	8.514*** (5.03)	8.514*** (5.03)	-2.548*** (-3.87)	-2.548*** (-3.87)
Log Assets _{f,j-1}	0.016 (1.32)	0.083** (2.49)	1.210** (1.98)	5.174*** (2.95)	-0.209 (-0.70)	-1.786** (-2.01)
Market-to-Book Ratio _{f,j-1}	0.054*** (6.66)	0.056*** (6.13)	3.112*** (7.56)	3.368*** (6.86)	-0.782*** (-4.13)	-0.716*** (-3.47)
Leverage _{f,j-1}	-0.038 (-0.62)	-0.033 (-0.61)	-1.851 (-0.63)	-0.323 (-0.07)	1.655 (1.08)	3.404 (1.00)
Investment/Sales _{f,j-1}	0.163*** (2.67)	0.110 (1.50)	12.139*** (3.85)	3.527 (1.08)	0.115 (0.08)	-0.864 (-0.53)
Analyst Coverage _{f,j}	-0.007*** (-3.42)	0.006** (2.11)	-0.322*** (-3.07)	0.387** (2.39)	0.065 (1.25)	0.058 (0.60)
Industry Fixed Effects	Yes	No	Yes	No	Yes	No
Fiscal-Year-End Fixed Effects	Yes	No	Yes	No	Yes	No
Firm Fixed Effects	No	Yes	No	Yes	No	Yes
Fiscal-Quarter Fixed Effects	No	Yes	No	Yes	No	Yes
Calendar-Quarter Fixed Effects	No	Yes	No	Yes	No	Yes
Obs.	11,820	11,820	11,820	11,820	11,820	11,820
Adj. R-sq.	0.095	0.031	0.105	0.034	0.050	0.010

The sample is 3,906 U.S. firms in the intersection of the Compustat and IBES databases. The sample excludes financials and utilities, firms that voluntarily adopted option expensing, and firms that changed their fiscal year-end in 2005 or 2006. Observations are at the firm-fiscal quarter level, for the four fiscal quarters before to four fiscal quarters after each firm's (staggered)

5.5 Table 4: analyst recommendations

Read:

- Analysts downgraded high-option firms after FAS 123-R.
- Analyst Rec falls by about 0.060 to 0.069 in FQ1 for high-option firms relative to low-option firms.
- The percentage of buy recommendations falls by around 2.65 to 3.35 percentage points.
- The percentage of sell recommendations rises by around 1.10 to 1.39 percentage points.
- Recommendation effects remain negative after FQ1, indicating analysts continue to view high-option firms as overvalued relative to low-option firms.

Economic message: Recommendation downgrades indicate that analysts do more than correct a mechanical EPS forecast. They update their valuation view of high-option firms.

5.6 Table 5: stock market valuations

Read:

- Market-to-book ratios fall for high-option firms after FAS 123-R.
- The average post-rule coefficient is about -0.160 to -0.140 in the first specifications.
- The quarter-by-quarter estimates show a widening valuation gap over time.
- By later post-compliance quarters, the market-to-book decline becomes substantially larger.
- The timing lines up with investors observing earnings misses and analyst updates.

Economic message: Investors also exhibit limited attention, but they appear to learn more slowly than analysts. Market valuations adjust gradually after option costs become salient.

Table 5
Effect of FAS 123-R on stock market valuations.

Dependent Variable Estimation Window	Market-to-Book Ratio _{<i>f,t</i>}			
	[-4;+4] Fiscal Quarters around FAS 123-R			
	(1)	(2)	(3)	(4)
<i>Post FAS 123-R</i> _{<i>f,t</i>} × <i>High-Option Firm</i> _{<i>f</i>}	-0.160*** (-3.33)	-0.140*** (-3.56)		
<i>Post FAS 123-R FQ1</i> _{<i>f,t</i>} × <i>High-Option Firm</i> _{<i>f</i>}			-0.025 (-0.53)	0.036 (0.89)
<i>Post FAS 123-R FQ2</i> _{<i>f,t</i>} × <i>High-Option Firm</i> _{<i>f</i>}			-0.141** (-2.39)	-0.135*** (-2.85)
<i>Post FAS 123-R FQ3</i> _{<i>f,t</i>} × <i>High-Option Firm</i> _{<i>f</i>}			-0.184*** (-3.12)	-0.198*** (-4.14)
<i>Post FAS 123-R FQ4</i> _{<i>f,t</i>} × <i>High-Option Firm</i> _{<i>f</i>}			-0.310*** (-5.15)	-0.298*** (-5.94)
<i>Post FAS 123-R</i> _{<i>f,t</i>}	0.069*** (4.15)	1.309*** (2.76)		
<i>Post FAS 123-R FQ1</i> _{<i>f,t</i>}			0.133*** (7.67)	1.179** (2.27)
<i>Post FAS 123-R FQ2</i> _{<i>f,t</i>}			0.090*** (4.27)	1.135** (2.19)
<i>Post FAS 123-R FQ3</i> _{<i>f,t</i>}			0.016 (0.79)	1.173** (2.27)
<i>Post FAS 123-R FQ4</i> _{<i>f,t</i>}			0.033* (1.65)	1.134** (2.20)
<i>High-Option Firm</i> _{<i>f</i>}	0.372*** (3.85)		0.375*** (3.89)	
<i>Log Assets</i> _{<i>f,t-1</i>}	-0.369*** (-10.22)	-0.815*** (-9.46)	-0.369*** (-10.21)	-0.798*** (-9.38)
<i>Leverage</i> _{<i>f,t-1</i>}	-0.700*** (-3.59)	-0.379*** (-2.04)	-0.690*** (-3.53)	-0.352* (-1.90)
<i>Investment/Sales</i> _{<i>f,t-1</i>}	1.183*** (3.88)	0.037 (0.18)	1.162*** (3.78)	0.013 (0.06)
<i>EBITDA/Sales</i> _{<i>f,t-1</i>}	2.701*** (4.70)	2.065*** (5.54)	2.646*** (4.59)	1.996*** (5.38)
<i>Analyst Coverage</i> _{<i>f,t</i>}	0.081*** (12.59)	0.011*** (3.02)	0.082*** (12.63)	0.012*** (3.13)
Industry Fixed Effects	Yes	No	Yes	No
Fiscal-Year-End Fixed Effects	Yes	No	Yes	No
Firm Fixed Effects	No	Yes	No	Yes
Fiscal-Quarter Fixed Effects	No	Yes	No	Yes
Calendar-Quarter Fixed Effects	No	Yes	No	Yes
<i>N</i>	10,720	10,720	10,720	10,720

5.7 Table 6 + Table 7: confounding channels and controls

Read:

- Table 6 tests whether FAS 123-R changed risk-taking, manipulation, and turnover outcomes for high-option firms.
- Some variables related to risk-taking or manipulation change, but many potential confounds do not.
- Table 7 then re-estimates the main outcomes while controlling for the confounding outcomes that changed after FAS 123-R.
- The main limited-attention effects remain: Missed Forecast rises, Analyst Rec falls, and Market-to-Book Ratio falls.
- In Table 7, the key coefficients remain statistically significant: Missed Forecast about 0.068, Analyst Rec about -0.062, and Market-to-Book about -0.100.

Economic message: The paper addresses the reuse-of-natural-experiments problem by explicitly identifying and controlling for other FAS 123-R channels.

Table 6

Effect of FAS 123-R on risk-taking, manipulation, and turnover outcomes.

Dependent Variable	Channel	Post FAS 123-R _{1,t} × High-Option Firm _{1,t}		Post FAS 123-R _{2,t}		High-Option Firm _{2,t}		Controls	Industry Fixed Effects	Fiscal-Year-End Fixed Effects	Obs.	Adj. R-sq.
		Coeff.	t-stat	Coeff.	t-stat	Coeff.	t-stat					
CF Volatility _{1,t}	RT	(1) 8.696***	(5.30)	2.158	(1.59)	48.652***	(9.31)	Yes	Yes	Yes	11,771	0.515
Book Leverage _{1,t}	RT	(2) 0.001	(0.26)	0.001	(0.34)	-0.071***	(-5.44)	Yes	Yes	Yes	11,787	0.282
Cash Holdings _{1,t}	RT	(3) -0.015***	(-2.75)	-0.000	(-0.02)	0.042***	(3.53)	Yes	Yes	Yes	11,789	0.364
Sales Concentration _{1,t}	RT	(4) 0.001	(0.10)	-0.000	(-0.03)	0.030*	(1.70)	Yes	Yes	Yes	2,681	0.285
Log Stock Volatility _{1,t}	RT	(5) -0.016	(-0.70)	0.043***	(3.37)	0.028	(0.83)	Yes	Yes	Yes	11,769	0.423
Debt Maturity _{1,t}	RT	(6) 0.033	(1.35)	0.004	(0.36)	-0.068**	(-2.19)	Yes	Yes	Yes	2,969	0.293
Accruals 1 _{1,t}	RT	(7) 0.014	(0.70)	0.005	(0.76)	-0.008	(-0.46)	Yes	Yes	Yes	11,537	0.413
Accruals 2 _{1,t}	RT	(8) 0.004	(0.24)	-0.009	(-1.45)	0.000	(0.00)	Yes	Yes	Yes	9,475	0.432
FX Exposure _{1,t}	RT	(9) -0.026	(-0.86)	0.268***	(16.05)	-0.004	(-0.15)	Yes	Yes	Yes	11,767	0.112
Share Repurchase _{1,t}	MA	(10) -0.598	(-1.25)	1.058***	(2.93)	2.152***	(3.23)	Yes	Yes	Yes	10,580	0.155
Dividend Payout _{1,t}	MA	(11) -0.002	(-0.21)	-0.005	(-0.95)	-0.046*	(-1.70)	Yes	Yes	Yes	11,789	0.331
Dis. SGAs _{1,t}	MA	(12) -0.015	(-1.40)	0.010	(1.55)	0.015	(1.40)	Yes	Yes	Yes	10,485	0.001
Dis. Accruals _{1,t}	MA	(13) 0.000	(0.02)	0.002***	(2.63)	-0.006*	(-1.90)	Yes	Yes	Yes	9,588	0.088
Dis. Asset Sales _{1,t}	MA	(14) -0.000	(-1.20)	-0.000**	(-2.56)	-0.000	(-0.30)	Yes	Yes	Yes	7,557	0.033
R&D/Capital _{1,t}	RT, MA	(15) 0.001	(1.02)	0.000	(0.44)	0.006***	(4.15)	Yes	Yes	Yes	11,066	0.503
Capex/Capital _{1,t}	RT, MA	(16) 0.005	(1.30)	-0.000	(-0.03)	-0.017	(-1.19)	Yes	Yes	Yes	11,059	0.528
CFO Turnover _{1,t}	TO	(17) 0.019	(0.69)	-0.042**	(-2.44)	-0.036	(-1.45)	Yes	Yes	Yes	2,820	0.010
Vol. CFO Turnover _{1,t}	TO	(18) 0.005	(0.19)	-0.027*	(-1.69)	-0.024	(-1.03)	Yes	Yes	Yes	2,820	0.002

This table examines the effect of FAS 123-R on outcome variables related to the risk-taking channel (RT), the manipulation channel (MA), and the turnover channel (TO). Which channel is tested is indicated in the table. The sample is U.S. firms in the intersection of the Compustat and IBES databases for which the outcome variables are available. The sample excludes financials and utilities, firms that voluntarily adopted option expensing, and firms that changed their fiscal year-end in 2005 or 2006. Observations are at the firm-fiscal quarter level, for the four fiscal quarters before to four fiscal quarters after each firm's (staggered) FAS 123-R compliance date (except in Rows 4, 6, 17, and 18, where we use outcome variables measured at the annual level). We label the analysis window as [-4;-4] fiscal quarters, with option expenses recorded starting in fiscal quarter 1. CF Volatility is the standard deviation of net cashflows from operating activities over the previous 8 fiscal quarters; Book Leverage is debt over assets; Cash Holdings is cash over assets; Sales Concentration is the sum of the square annual sales of a firm's segments divided by the square of the firm's annual total sales; Log Stock Volatility is the natural logarithm of the variance of daily stock returns over the 230 trading days before the end of the fiscal quarter; Debt Maturity is long-term debt over total debt; Accruals 1 and Accruals 2 are measures of accruals; FX Exposure is a trade-weighted index of the real bilateral exchange rates between the U.S. dollar and the major currencies that trade freely outside of their country of issue

Table 6: possible confounding channels.

Table 7

Effect of FAS 123-R: Controlling for risk-taking and manipulation outcomes.

Dependent Variable	Missed Forecast _{1,t}	Analyst Rec _{1,t}	Market-to-Book Ratio _{1,t}
Estimation Window	[-4;-4] Fiscal Quarters around FAS 123-R		
	(1)	(2)	(3)
Post FAS 123-R _{1,t} × High-Option Firm _{1,t}	0.068*** (3.54)	-0.062*** (-3.14)	-0.100** (-2.01)
Post FAS 123-R _{2,t}	-0.004 (-0.38)	-0.011 (-1.01)	0.062*** (3.77)
High-Option Firm _{1,t}	-0.045* (-1.65)	0.221*** (4.84)	0.087 (0.72)
Log Assets _{1,t-1}	-0.059*** (-7.36)	-0.001 (-0.04)	-0.350*** (-7.79)
Market-to-Book Ratio _{1,t-1}	-0.015*** (-3.15)	0.060*** (7.42)	
Leverage _{1,t-1}	0.108*** (3.08)	-0.052 (-0.86)	-0.287 (-1.56)
Investment/Sales _{1,t-1}	0.103*** (2.86)	0.220*** (3.47)	0.866*** (2.96)
EBITDA/Sales _{1,t-1}			3.428*** (6.09)
Analyst Coverage _{1,t}	0.002 (1.26)	-0.005** (-2.32)	0.068*** (11.01)
CF Volatility _{1,t} × High-Option Firm _{1,t}	-0.000 (-0.32)	-0.006*** (-3.85)	0.005 (1.00)
Cash Holdings _{1,t} × High-Option Firm _{1,t}	0.110* (1.95)	0.065 (0.61)	0.298 (0.85)
CF Volatility _{1,t}	0.000*** (3.27)	0.000 (0.67)	0.001*** (3.25)
Cash Holdings _{1,t}	-0.250*** (-5.28)	-0.352*** (-3.71)	1.650*** (5.84)
Corresponding Estimate: Same Sample, w/o Add. Controls			
Post FAS 123-R _{1,t} × High-Option Firm _{1,t}	0.066*** (3.49)	-0.066*** (-3.35)	-0.161*** (-3.33)
Industry Fixed Effects	Yes	Yes	Yes
Fiscal-Year-End Fixed Effects	Yes	Yes	Yes

Table 7: main effects with confound controls.

5.8 Structural model panel + Table 8: SMM estimation and counterfactuals

Read:

- The structural model incorporates limited attention, compensation design, manipulation, and risk-taking incentives.
- The estimated attention parameter is highest in the non-expensing regime, lower in the interim regime, and much lower in the expensing regime.
- Panel A of Table 8 estimates $\alpha_N = 0.086$, $\alpha_{E-} = 0.062$, and $\alpha_E = 0.017$, all statistically significant.
- Panel B compares moments from the data with moments simulated by the model.
- Panel C conducts counterfactual exercises that isolate the effects of limited attention, manipulation incentives, and risk-taking incentives.

Economic message: The structural model quantifies the mechanism. It supports the interpretation that the valuation change is primarily due to increased attention to option costs rather than only simultaneous changes in firm behavior.

; compensation incentives.¹⁷ We also follow MV's framework since optimal contract may encourage some manipulation, and this does arise in other dynamic CEO pay models (e.g., Edmans et al., 2012). The setup includes a CEO (agent) and a board (principal) who work a firm owned by investors. In each period t , the CEO chooses effort risk σ_t and manipulation m_t . The expected cashflow process is:

$$c_t = (a_N \sigma_t + m_t \sigma_t - \theta M_t \sigma_t + \varpi \sigma_t^2) dt + \sigma_t d B_t,$$

where B_t is a standard Brownian motion. Risk-taking incentives are added in a similar spirit to Song (1995) (for tractability, the mean and variance of the cashflow process scale with the level of risk). When CEO chooses higher σ_t , she increases the expected value of cashflows but also amplifies the diffusion process.¹⁸ The marginal effect risk on (volatility-scaled) cashflows is ϖ . Following MV, M_t is the risk of accumulated manipulation between time 0 and t , and it reduces cashflows at the marginal rate θ . In this formulation, the CEO can boost cashflows by increasing m_t , but in future times $t' > t$ this reduces cashflows $dY_{t'}$ through a build-up in $M_{t'}$. This stock depreciates rate κ , so recent manipulation choices have more impact than those made in the distant past.

The CEO has exponential utility that increases with consumption c_t and decreases with cost functions for a_N , m_t , and σ_t . Following MV, the CEO works until a finite retirement date T , but continues to be paid until time $T + \tau$, where $\tau \geq 0$ is the length of time during which the firm can enforce compensation clawbacks. She can also save privately, which means that the CEO consumes the full compensation provided by the contract (compensation is equal to c_t).

The model incorporates FAS 123-R by specifying that the firm is in one of three states, denoted as $g_t = \{N, E^-, E^+\}$: N is the non-expensing state, E^- is the expensing state, and E^+ is the interim state.

which equals the expected value of cashflows net of CEO pay, both discounted at rate r . One motivation for the board's objective function is that a CEO typically sells equity as soon as it vests, so its market value at that time determines her consumption. The contract accounts for CEO incentive compatibility by eliciting optimal choices of a_t , m_t , and σ_t :

$$V(c, a, m, \sigma, g) = \mathbb{E}^{(a, m, \sigma)} \left[\int_0^\infty e^{-rt} \underbrace{(dY_t - c_t(1 - a_N)) 1_{E^+ T+t}}_{\text{Cashflows net of CEO pay}} dt \right] - \underbrace{e^{-r(T+\tau)} \frac{rT+\tau}{r} (1 - a_N)}_{\text{Post-retirement pay}}$$

As in MV, the optimal contract has a general form. The CEO receives a continuous stream of payments which is sensitive to cashflow shock realizations. In the presence of manipulation, the optimal contract is nonlinear. It conveys both short- and long-term incentives, which are determined by the sensitivity of CEO payments to firm performance over different horizons. Overall, the contract resembles equity pay with time-varying vesting periods.

The model can only be solved numerically.²¹ In brief, we solve for a stochastic control problem that requires keeping track of two state variables: i) The CEO's continuation utility W_t , which is the expected discounted utility that the contract provides the CEO from t to $T + \tau$; and ii) the contract's long-term incentives μ_t , which are given by the expected discounted future sensitivity of W_t to cashflows (the contract's future pay-performance sensitivity, PPS). CARA preferences, the possibility of private savings, and the scalability of the cashflow process with

Original structural-model equation panel.

5.9 Table 9 + Table 10: analyst learning

Read:

- Table 9 studies domain-specific analyst learning.
- Analysts who had prior exposure to voluntary adopters or early compliers appear better able to process option costs.
- High-option firms followed by analysts with less relevant prior coverage experience larger forecast-miss effects.
- Table 10 studies domain-general analyst learning using previous analyst experience.
- The evidence suggests that both task-specific experience and broader analyst experience matter for learning.

Economic message: Limited attention is not fixed. Analysts learn from prior exposure, and experience helps them process the same information more accurately when reporting salience changes.

Table 8
Structural estimation: SMM estimation results.

Panel A: Parameter estimates								
Parameter	a_N	a_E^-	a_E^+	$\zeta_{N,E}$	l	κ	θ	ϖ
Value	0.086*** (0.000005)	0.062*** (0.000001)	0.017*** (0.000002)	0.471*** (0.000001)	5.148*** (0.00017)	0.111*** (0.000004)	0.130*** (0.000001)	0.161*** (0.000005)
Panel B: Moments								
Moment Specification				Actual		Model		
$\beta^{Treated \times I^E}$				-0.091		-0.084		
$\beta^{M/B \text{ Ratio}}$				-0.001		0.000		
$\beta^{CF \text{ Volatility}}$				0.007		-0.003		
β^{I^E}				0.011		0.000		
$\beta^{Treated \times I^E}$				-0.169		-0.160		
$\beta^{Treated \times I^{E^+}}$				0.283		-0.027		
$\beta^{Treated \times I^{E^-}}$				-0.003		-0.001		
$\beta^{CF \text{ Volatility}}$				0.002		-0.001		
β^{I^E}				0.014		0.000		
$\beta^{Treated \times I^{E^+}}$				-0.225		-0.033		
$\beta^{Intercept}$				0.109		0.036		
$\beta^{M/B \text{ Ratio}}$				0.001		0.000		
$\beta^{CF \text{ Volatility}}$				-0.002		0.000		
β^{I^E}				0.126		0.067		
$\beta^{I^{E^+}}$				0.543		0.198		
Cov(Y_t, s_t)				-0.001		0.001		
Cov(M/B Ratio, CF Volatility)				0.001		0.000		
Test of Overidentifying Restrictions								
χ^2						5,092		
p-value						0.000		

This table reports parameter estimates from the SMM procedure (Panel A) and simulated moments from the model (Panel B). In Panel A, a_N , a_E^- , and a_E^+ are the limited attention parameters in the non-expensing state (N), interim state (E^-), and expensing state (E^+), respectively. $\zeta_{N,E}$ denotes the transition intensity from the non-expensing state (N) to the interim state (E^-). l represents the personal cost of manipulation, κ denotes the rate of depreciation of the stock of manipulation, θ captures the marginal effect of manipulation on cashflows, and ϖ captures the marginal effect of volatility on cashflows. In Panel B, regression coefficients that serve as moments are denoted by β . The dependent variables are indicated as subscripts in each β coefficient, while the independent variables are indicated with superscripts. Dependent variables in these regressions include the market-to-book ratio (*M/B Ratio*), cashflow volatility (*CF Volatility*), the vesting of incentives (Δz), operating cashflows (Y), and the duration of CEO incentives (z). The key independent variables include interaction terms between *Treated* and indicators for the expensing state (I^E) and for the interim state (I^E). In the actual data,

Table 8: SMM estimation results.

Table 9

Effects of domain-specific analyst learning.

Panel A: Firms with fiscal years ending in June-Sept					Panel B: Firms with fiscal years ending in Oct-Dec				
Dependent Variable	Missed Forecast _{<i>j,t</i>}		Analyst Rec _{<i>j,t</i>}		Missed Forecast _{<i>j,t</i>}		Analyst Rec _{<i>j,t</i>}		
Sample	Previous Analyst Coverage of Voluntary Adopters				Previous Analyst Coverage of Early Compliers				
Estimation Window	High Coverage	Low Coverage	High Coverage	Low Coverage	High Coverage	Low Coverage	High Coverage	Low Coverage	
	[-4:+4] Fiscal Quarters around FAS 123-R				[-4:+4] Fiscal Quarters around FAS 123-R				
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	
Post FAS 123-R _{<i>j,t</i>} × High-Option Firm _{<i>j</i>}	-0.065 (-0.67)	0.281*** (4.07)	0.017 (0.20)	-0.139* (-1.86)	0.017 (0.51)	0.071** (2.31)	-0.054* (-1.67)	-0.118*** (-3.76)	
Post FAS 123-R _{<i>j,t</i>}	-0.018 (-0.47)	-0.106** (-2.17)	-0.067** (-2.21)	0.001 (0.03)	0.008 (0.41)	-0.005 (-0.29)	0.007 (0.27)	-0.010 (-0.52)	
High-Option Firm _{<i>j</i>}	0.016 (0.20)	-0.217 (-3.11)	0.151 (1.22)	0.053 (0.80)	-0.019 (1.02)	0.148*** (4.61)	0.114** (2.99)	0.114** (2.98)	
Log Assets _{<i>j,t-1</i>}	-0.032 (-1.08)	-0.052 (-1.62)	0.029 (1.14)	-0.029 (-0.54)	-0.044*** (-4.27)	0.033* (2.33)	-0.015 (-0.76)	-0.015 (-0.76)	
Market-to-Book Ratio _{<i>j,t-1</i>}	-0.049** (-2.18)	0.006 (0.20)	0.002** (2.44)	0.124*** (3.40)	-0.024*** (-3.06)	-0.017** (-2.32)	0.044*** (3.12)	0.050*** (4.78)	
Leverage _{<i>j,t-1</i>}	0.007 (0.51)	0.172 (1.02)	-0.427* (-1.86)	0.219 (0.71)	0.096 (1.43)	0.103** (2.08)	0.078 (0.86)	-0.012 (-0.14)	
Investment/Sales _{<i>j,t-1</i>}	-0.593 (-4.57)	-0.222** (-2.54)	0.957 (0.89)	-0.478 (-1.33)	0.124 (0.96)	0.065 (1.54)	-0.054 (-0.38)	0.211*** (3.38)	
Analyst Coverage _{<i>j,t</i>}	0.000 (0.03)	-0.006 (-1.43)	-0.015** (-2.25)	-0.021*** (-2.37)	0.000 (0.14)	0.000 (0.05)	-0.006* (-1.87)	-0.001 (-0.39)	
Industry Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	
Fiscal-Year-End Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	
Obs.	827	835	825	833	3,955	4,430	3,937	4,413	
Adj. R-sq.	0.014	0.041	0.202	0.163	0.030	0.032	0.090	0.155	

In Panel A, the sample is 605 firms with a fiscal year ending in June to September. For each firm, we measure the number of analysts who previously covered more than one voluntary adopter, divided by the total number of analysts covering the firm when it began to comply with FAS 123-R. (Voluntary adopters are firms that voluntarily adopted stock option expensing prior to FAS 123-R compliance.) Firms are split into high coverage and low coverage groups, depending on whether this ratio is above or below the sample median. In Panel B, the sample is 2,811 firms with a fiscal year ending in October to December. For each firm, we calculate the number of analysts who previously covered more than one early complier, divided by the total number of analysts covering the firm when it began to comply with FAS 123-R. Firms are split into high and low coverage groups depending on whether this ratio is above or below the sample median.

Table 9: domain-specific analyst learning.

Table 10

Effects of domain-general analyst learning.

Dependent Variable	Missed Forecast $f_{j,t}$		Analyst Rec $f_{j,t}$	
	Previous Analyst Experience		Previous Analyst Experience	
Sample	High Coverage	Low Coverage	High Coverage	Low Coverage
Estimation Window	[-4:+4] Fiscal Quarters around FAS 123-R		[-4:+4] Fiscal Quarters around FAS 123-R	
	(1)	(2)	(3)	(4)
Post FAS 123-R _{j,t} × High-Option Firm _j	0.033 (1.21)	0.097*** (3.56)	-0.039 (-1.42)	-0.098*** (-3.49)
Post FAS 123-R _{j,t}	0.023 (1.34)	-0.026* (-1.75)	-0.028* (-1.83)	0.017 (1.10)
High-Option Firm _j	0.003 (0.10)	-0.037 (-1.35)	0.129*** (2.68)	0.189*** (4.20)
Log Assets _{j,t-1}	-0.035*** (-3.93)	-0.035*** (-4.13)	0.014 (0.85)	0.018 (1.07)
Market-to-Book Ratio _{j,t-1}	0.136** (2.40)	0.114** (2.45)	0.101 (1.15)	-0.112 (-1.29)
Leverage _{j,t-1}	0.070 (1.37)	0.058 (1.09)	0.204*** (3.00)	0.137 (1.41)
Investment/Sales _{j,t-1}	-0.020*** (-2.98)	-0.010 (-1.46)	0.060*** (3.03)	0.049*** (4.26)
Analyst Coverage _{j,t}	0.000 (0.22)	-0.001 (-0.40)	-0.006** (-2.02)	-0.009*** (-3.14)
Industry Fixed Effects	Yes	Yes	Yes	Yes
Fiscal-Year-End Fixed Effects	Yes	Yes	Yes	Yes
Obs.	5,694	5,983	5,671	5,953
Adj. R-sq.	0.035	0.033	0.112	0.112

The sample is 3,906 U.S. firms in the intersection of the Compustat and IBES databases. For each firm, we measure the number of years between the median analyst's first earnings forecast or stock recommendation in IBES and the analyst's last forecast or recommendation for the firm before it complied with FAS 123-R. Firms are split into high and low coverage groups depending on whether this value is above or below the sample median. The sample excludes financials and utilities, firms that voluntarily adopted option expensing, and firms that changed their fiscal year-end in 2005 or 2006. Observations are at the firm-fiscal quarter level, for the four fiscal quarters before to four fiscal quarters after each firm's (staggered) FAS 123-R compliance date. We label the analysis window as [-4:+4] fiscal quarters, with option expenses recorded starting in fiscal quarter 1. Regressions are restricted to firms that are followed by at least three analysts and that are in the sample both before and after

Table 10: domain-general analyst learning.

5.10 Table 11 + Table 12: investor learning and option-related return dynamics

Read:

- Table 11 studies market reactions to missed and met earnings forecasts.
- High-option firms that miss forecasts experience negative returns around the earnings announcement and continued negative drift after the announcement.
- The post-announcement drift suggests investors did not immediately incorporate the full implications of option costs.
- Table 12 compares option-related return dynamics in the non-expensing and expensing regimes.
- In the non-expensing regime, high-option firms experience return dynamics consistent with investors underweighting option costs.
- In the expensing regime, the same option-related misvaluation dynamics are substantially reduced.

Economic message: The market learns slowly. Analysts' updated forecasts and recommendations help, but investors continue to adjust after observing option-related earnings surprises.

Table 11

Investor learning: Market reactions to missed and met earnings forecasts.

Panel A: Announcement of EPS for FQ1 that miss forecast			
Estimation Window (Trading Days around Announcement):	Pre-Announcement Period [-20,-4]	Announcement Period [-3,+3]	Post-Announcement Period [+4,+60]
High-Option Firms	-1.72%	-3.72%	-6.27%
Low-Option Firms	-1.12%	-2.95%	-0.33%
Δ High-Option vs. Low-Option Firms	-0.60% (-0.82)	-0.77% (-1.16)	-5.94% (-3.95)***
Obs.	828	828	815
Panel B: Announcement of EPS for FQ1 that meet forecast			
Estimation Window (Trading Days around Announcement):	Pre-Announcement Period [-20,-4]	Announcement Period [-3,+3]	Post-Announcement Period [+4,+60]
High-Option Firms	0.78%	0.95%	-5.12%
Low-Option Firms	0.36%	1.45%	-0.49%
Δ High-Option vs. Low-Option Firms	0.42% (0.66)	-0.50% (-0.79)	-4.63% (-3.32)***
Obs.	882	882	878

Table 11: investor learning from EPS announcements.

Table 12

Option-related return dynamics: Non-expensing vs. expensing regime.

Dependent Variable	Panel A: High-option firms		Panel B: Low-option firms	
	EA CAR _{j,t} [-12:-1] Fiscal Quarters before FAS 123-R	EA CAR _{j,t} [+1:+8] Fiscal Quarters after FAS 123-R	EA CAR _{j,t} [-12:-1] Fiscal Quarters before FAS 123-R	EA CAR _{j,t} [+1:+8] Fiscal Quarters after FAS 123-R
Estimation Window	(1)	(2)	(3)	(4)
Δ Shares Out _{j,t}	-0.092*** (-5.24)	0.024 (1.26)	-0.051*** (-3.25)	-0.019 (-0.72)
Log Assets _{j,t-1}	-0.004** (-2.47)	0.004*** (3.39)	0.001* (1.66)	0.002** (2.52)
Market-to-Book Ratio _{j,t-1}	-0.007*** (-7.75)	-0.001 (-1.40)	-0.003*** (-3.27)	-0.001 (-0.94)
Leverage _{j,t-1}	0.013 (1.24)	-0.012 (-1.23)	0.002 (0.37)	-0.017** (-2.28)
Investment/Sales _{j,t-1}	-0.008 (-1.14)	-0.005 (-0.84)	0.004 (0.41)	-0.011 (-1.06)
Shares Repurchased _{j,t}	0.644*** (3.49)	0.402** (2.29)	0.103 (1.12)	0.247*** (2.59)
Convertible Debt/Assets _{j,t-1}	-0.011 (-0.65)	-0.010 (-0.65)	-0.028** (-2.25)	0.022 (1.39)
Stock Return _{j,t-1}	-0.023*** (-4.63)	-0.020*** (-3.11)	-0.022*** (-4.35)	-0.023*** (-2.96)
Industry Fixed Effects	Yes	Yes	Yes	Yes
Fiscal-Year-End Fixed Effects	Yes	Yes	Yes	Yes
Obs.	12,734	8,889	13,858	8,794
Adj. R-sq.	0.016	0.006	0.007	0.008

Panel A examines the market reaction to changes in shares outstanding among high-option firms, while Panel B examines the reaction among low-option firms. High-option firms are firms with an above-median ratio of *Fair Value of Options Granted/Total Assets* in either fiscal year 2004 or fiscal year 2005, while all other firms are low-option firms. The sample is 3,906 U.S. firms in the intersection of the Compustat and IBES databases. The sample excludes financials and utilities, firms that voluntarily adopted

Table 12: option-related return dynamics.

6 Short summary

- The paper studies whether sophisticated market participants pay limited attention to detailed financial statement information.
- FAS 123-R moved option compensation costs from footnotes into the income statement.
- The rule increased the visibility of option costs without changing their economic magnitude.
- High-option firms experienced lower reported EPS after FAS 123-R.
- Analysts were more likely to miss EPS forecasts for high-option firms in the first two quarters after compliance.
- Forecast accuracy improved by later quarters, consistent with analyst learning.
- Analysts downgraded high-option firms, and recommendation effects persisted.
- Investors gradually marked down high-option firms' market-to-book ratios.
- The authors address confounding channels from prior FAS 123-R papers using reduced-form channel tests.
- They then estimate a structural model to quantify limited attention.
- Structural estimates imply that attention to option costs rises substantially after expensing.
- The paper contributes both to limited-attention research and to empirical methods for reusing natural experiments.

7 Conclusion

7.1 Core conclusion

Cronqvist, Ladika, Pazaj, and Sautner show that financial statement presentation can affect even sophisticated market participants. When FAS 123-R made option compensation expenses more visible, analysts and investors revised forecasts, recommendations, and valuations for high-option firms.

7.2 Economic intuition

Information is not equally processed in all formats. Before FAS 123-R, option costs were available but buried in footnotes, so analysts and investors with limited attention underweighted them. After expensing, the same costs became salient in earnings, causing forecast misses, downgrades, and valuation declines.

7.3 Methodological takeaway

The paper also shows how to reuse a natural experiment responsibly. Because FAS 123-R affected many firm choices, the authors combine reduced-form tests of confounding channels with structural estimation. This approach helps separate the limited-attention mechanism from simultaneous changes in incentives, risk-taking, or manipulation.

7.4 Takeaway

The enduring lesson is that detail matters, but presentation of detail matters too. Financial markets may not fully process information unless it is sufficiently salient, even when the information is technically disclosed and the relevant agents are professionals.